

THERMODYNAMICS SOLUTIONS

Exercise: 1

IAT-Based problems

1.

$$\Delta H = \Delta H_f + \Delta H_v = 80 \times 18 + 596 \times 18$$

$$\Delta H = 12168 \text{ cal/mol}$$

$\Delta H = \Delta U + P\Delta V$ –according to the first law of thermodynamics

Now $PV = nRT$

$$P\Delta V = nRT = 1 \times 8 - 314 \times \frac{278}{0.18} = 543 \text{ cal}$$

$$\Delta U = \Delta H - P\Delta V$$

$$\Delta U = 12168 - 543 \text{ cal}$$

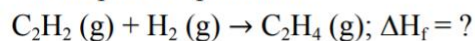
$$\Delta U = 11625 \text{ cal}$$

Hence, ΔH is 12168 cal, and ΔU is 11625 cal.

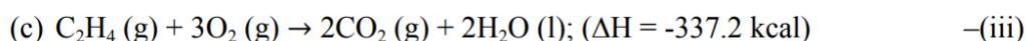
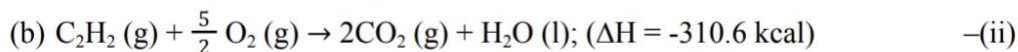
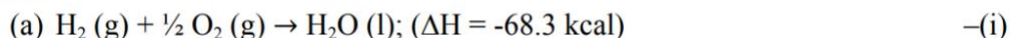
Therefore, the correct option is A.

2.

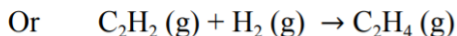
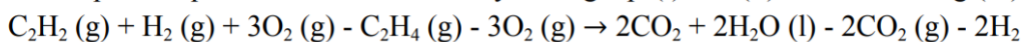
The required equation is



Given



The required equation can be achieved by adding eqs (i) and (ii) and subtracting (iii)



$$\Delta H = -68.3 - 310.6 - (-337.2) = -378 + 337.2 = -41.7 \text{ kcal}$$

We know that

$$\Delta H = \Delta U + \Delta nRT$$

$$\text{Or } \Delta U = \Delta H - \Delta nRT$$

$$\text{Or } \Delta n = (1-2) = -1; \quad R = 2 \times 10^{-3} \text{ kcalmol}^{-1} \text{ K}^{-1}$$

$$\text{And } T = (25 + 273) = 298 \text{ K}$$

Substituting the values in above equation

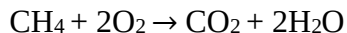
$$\Delta U = -41.7 - (-1)(2 \times 10^{-3})(298)$$

$$\Delta U = -41.104 \text{ kcal}$$

Therefore, the correct option is C.

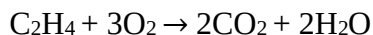
3.

Combustion of CH_4 –



1 vol 1 vol

Combustion of C_2H_4 –



1 vol 2 vol

Let V litre be the volume of CH_4 in the mixture

Total volume of mixture = 3.67 L

$\therefore$ The volume of C_2H_4 in the mixture = $(3.67 - V)$ L

$\therefore$ Volume of CO_2 formed during combustion of C_2H_4 = $V + 2(3.67 - V)$

However, given that the volume of CO_2 formed during the combustion of C_2H_4 is 6.11 L

$$\therefore V + 2(3.67 - V) = 6.11$$

$$\Rightarrow V = 1.23 \text{ L}$$

Therefore,

$$\text{Volume of } \text{CH}_4 \text{ in 1 L of mixture} = \frac{1.23}{3.67}$$

$$\text{Volume of } \text{C}_2\text{H}_4 \text{ in the mixture} = \frac{2.44}{3.67}$$

As we know that volume of 1 mole of any gas at 25°C and

$$1 \text{ atm} = \frac{22.4}{273} \times 298 = 24.45 \text{ L [by using Charle's law]}$$

Given that –

Heat of combustion of methane = -891 kJ/mol

Heat of combustion of ethylene = -1423 kJ/mol

$$\text{Heat evolved due to the combustion of } \text{CH}_4 = \frac{1.23 \times (-891)}{3.67 \times 24.45} = -12.213 \text{ KJ}$$

$$\text{Heat evolved due to the combustion of } \text{C}_2\text{H}_4 = \frac{2.44 \times (-1423)}{3.67 \times 24.45} = -38.694 \text{ KJ}$$

$$\therefore \text{Total heat evolved} = (12.213) + (-38.694) = -50.907 \text{ KJ}$$

Therefore, the correct option is A.

4.

Heat supplied, $q = 500 \text{ J}$

The temperature of the system is raised from 298 to 308 K

For this process, volume is constant, i.e., $\Delta V = 0$

Work done is given by,

$$\Rightarrow W = -P\Delta V$$

$$= 0$$

Now, applying the first law of thermodynamics and using $W = 0$, we got:

$$\Delta U = Q + W$$

$$= q + 0$$

$$= q$$

So we got $Q = \Delta U = 500\text{J}$ and $W = 0$.

Therefore, the correct option is B.

5. Internal energy is the sum of microscopic forms of energy within a system, including the kinetic and potential energy of particles. It represents the total energy content at the molecular level. Changes occur through processes like heating or cooling, and it follows the first law of thermodynamics. Therefore, internal energy includes translational, rotational, vibrational, electronic and nuclear energies. It does not include gravitational pull.

Therefore, the correct option is D.

6.

We know that, $\Delta U = Q + W$

If ΔU is positive, the internal energy of the system will increase. For this to happen, the system must have gained more energy through heat (Q) than it has lost through work done on the surroundings (W). In other words, the work done by the system is performed on its surroundings, resulting in a negative contribution to the internal energy equation.

Therefore, for ΔU to be positive, W must be negative.

Therefore, the correct option is C.

7.

$n = 1$ mole

Pressure = 1 bar

Temperature = 100°C

$T = 100 + 273\text{K}$

$T = 373\text{K}$

$\Delta H = 41 \text{ kJ/mol}$

$R = 8.314 \times 10^{-3} \times \text{kJ/mol}$

$$\Delta H = \Delta U + \Delta n_g RT$$

$$\Delta U = \Delta H - \Delta n_g RT$$

$$\Delta U = 41 - (1 \times 8.314 \times 10^{-3} \times 373)$$

$$\Delta U = 37.904 \text{ kJ/mol}$$

Therefore, the correct option is A.

8.

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$P_1 = 1.013 \text{ MPa}$$

$$V_1 = 1 \text{ dm}^3$$

$$P_2 = 0.1013 \text{ MPa}$$

$$V_2 = ?$$

$$(1.013) \times (1)^{1.66} = (0.1013) \times (V_2)^{1.66}$$

$$V_2 = 10^{\frac{1}{1.66}} \text{ dm}^3 = 3.98 \approx 4 \text{ dm}^3$$

$$PV = nRT$$

$$\Rightarrow 1.013 \times 10^6 \times 1 \text{ dm}^3$$

$$\Rightarrow 0.4 \text{ mol} \times 25/3 \text{ J/mol/K} \times T$$

$$\Rightarrow T = 304 \text{ K}$$

$$T_f = T_i \left(\frac{V_i}{V_f} \right)^{\gamma-1}$$

$$T_f = 304 \left(\frac{1}{4} \right)^{(1.66-1)}$$

$$= 120.6 \text{ K}$$

$$W = \frac{(P_2 V_2 - P_1 V_1)}{(1-\gamma)}$$

$$W = \frac{(1.013 \times 10^5 \text{ Pa} \times 4 \times 10^{-3} \text{ m}^3 - 1.013 \times 10^6 \times 1 \times 10^{-3} \text{ m}^3)}{(1-1.66)}$$

$$W = \frac{6 \times 1.013 \times 10^2}{0.66}$$

Therefore, the correct option is B.

9.

Liquid x volume = 100 litres

$\Delta V = 2$ litres, $\Delta Q = \Delta H$, $P_{\text{ext}} = 10 \text{ atm}$

$$W = -P_{\text{ext}}(V_2 - V_1)$$

$$W = -1(2) \text{ L atm}$$

$$W = -2 \times 101.3 \text{ J}$$

$$W = -202.6 \text{ J}$$

Heat supplied, given = 202.6 J

$$\Delta Q = \Delta H$$

By the first law of thermodynamics

$$\Delta U = Q + W$$

$$\Delta U = 202.6 - 202.6$$

$$\Delta U = 0$$

Therefore, the correct option is D.

10. Work done during expansion is equal to the heat supplied to water to raise the temperature.

$$P(V_2 - V_1) = mS\Delta T$$

$$3 \times 2 \times 101.3 = 180 \times 4.18(T_F - 290)$$

$$T_f = 290.8 \text{ K}$$

Therefore, the correct option is B.

11. The enthalpy change for the reaction is the difference between the total enthalpy of products and the total enthalpy of reactants.

The correct option is C.

12.

The general formula of entropy changes

$$\Delta S = C_v \ln\left(\frac{T_2}{T_1}\right) + R \ln\left(\frac{V_2}{V_1}\right)$$

$$T_1 = 25^\circ\text{C} = 298 \text{ K}$$

$$T_2 = 100^\circ\text{C} = 373 \text{ K}$$

$$C_v = 5 \text{ cal}$$

$$R = 2 \text{ cal/mol K}$$

$$V_1 = 1 \text{ L}$$

$$V_2 = 10 \text{ L}$$

$$\Delta S = 5 \ln\left(\frac{373}{298}\right) + 2 \ln\left(\frac{10}{1}\right)$$

$$\Delta S = 5 \ln\left(\frac{373}{298}\right) + 2 \ln(10)$$

Therefore, the correct option is B.

13.

Using,

$$\Delta S_{transition} = \frac{\Delta H_{transition}}{T_{transition}}$$

$$\Delta S_{surr} = \frac{-\Delta H_{trans}}{T}$$

$$\Delta S_{trans} = \frac{-401.7}{368}$$

$$\Delta S_{surr} = \frac{+401.7}{273}$$

∴ temperature of ice bath = 273 K

$$\Delta S_{total} = \frac{-401.7}{368} + \frac{401.7}{273} = 0.38 \text{ J/K}$$

Therefore, the correct option is C.

14.

$$\Delta H = H_f^0(\text{products}) - H_f^0(\text{reactants})$$

The reactants are CO₂ H₂, and the products are CO and H₂O.

Hence, $\Delta H = -100.5 + (-241.8) - (-393.5 + 0) = 41.2 \text{ kJ}$

Therefore, the correct option is B.

15.

The work obtained by a reversible engine operating between two temperatures is calculated by multiplying the heat absorbed at the higher temperature with the number of cycles.

In this case, the engine operates between 1000K and 300K, and the heat absorbed at the higher temperature is 2.1kJ/cycle. The work obtained in 1000 cycles is, therefore, 2100 kJ.

Work obtained = 2.1 kJ/cycle × 1000 cycles

$$= 2100 \text{ kJ}$$

To calculate the heat absorbed at the higher temperature:

Heat absorbed = 2.1 kJ/cycle

The work obtained in 1000 cycles:

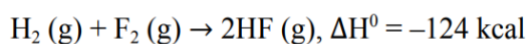
$$\begin{aligned}\text{Work obtained} &= \text{Heat absorbed} \times \text{Number of cycles} \\ &= 2.1 \text{ kJ/cycle} \times 1000 \text{ cycles} \\ &= 2100 \text{ kJ}\end{aligned}$$

The work obtained in a reversible engine operating between two temperatures is the product of the heat absorbed at the higher temperature and the number of cycles.

Therefore, the correct option is A.

16.

For the reaction,



$$\Delta H^0 = \sum BE(\text{Reactants}) - \sum BE(\text{Products})$$

$$-124 = \Delta H_{\text{H-H}} + \Delta H_{\text{F-F}} - 2\Delta H_{\text{H-F}}$$

$$-124 = 104 + 37.8 - 2\Delta H_{\text{H-F}}$$

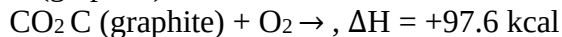
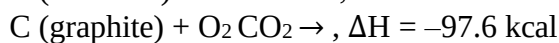
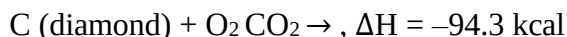
$$2\Delta H_{\text{H-F}} = 104 + 37.8 + 124 = 265.8 \text{ kcal}$$

$$\text{Bond energy of H-F} = \frac{265.8}{2} \text{ kcal} = 132.9 \text{ kcal}$$

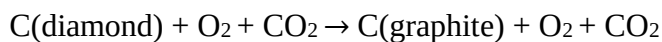
$$\therefore \Delta H^0 \text{ for the given reaction} = -132.9 \text{ kcal}$$

Therefore, the correct option is B.

17.



On adding both reactions:



$$\Delta H = -94.3 + 97.6 = 3.3 \text{ kcal/mol}$$

Therefore, the correct option is B.

18.

The greater the heat of neutralization, the stronger the acid to the more acidic strength. So, the acidic character order is $A > D > C > B$.

The greater the enthalpy of ionization of any acid or base, the weaker will be the acid or base. The acidic strength of an acid is proportional to the magnitude $\Delta_{\text{neut}}H$.

Therefore, the correct option is B.

19.

The enthalpy change for the reaction is the difference between the total enthalpy of products and the total enthalpy of reactants.

$$\Delta H_r = \left[4(\Delta H_f)_{CO_2} + 2(\Delta H_f)_{H_2O} - 2(\Delta H_f)_{C_2H_2} \right]$$

$$- 2601 = - 4(394) - 2(285.8) - (\Delta H_f)_{C_2H_2}$$

$$(\Delta H_f)_{C_2H_2} = 226.7$$

Hence, the heat of formation of $C_2H_2(g)$ is 226.7 kJ.

The enthalpies of the formation of elements in their standard states are considered to be zero. Hence, the enthalpy of $O_2(g)$ does not appear in the above expression.

Therefore, the correct option is D.

20.

$$m_{Zn} \cdot S_{Zn}(T_f - T_i) + m_{H_2O} \cdot S_{H_2O}(T_f - T_i) = 0$$

$$\Rightarrow (65.38g)(0.4J/g^{\circ}C)(T_f - 20^{\circ}C) + 180g(0.4J/g^{\circ}C) \times (T_f - 100^{\circ}C) = 0$$

$$\Rightarrow [(65.38)(0.4) + 180(420)]T_f = (65.38)(0.4)(20) + 180(420)(100)$$

$$\Rightarrow T_f = \left[\frac{(65.38)(0.4)(20) + 180(420)(100)}{(65.38)(0.4) + 180(420)} \right] = 97.3^{\circ}C$$

Therefore, the correct option is A.

21.

The internal energy (U) of an ideal gas is given by the formula:

$$U = n \times C_v \times T$$

where:

n is the number of moles,

C_v is the specific heat capacity at constant volume, and

T is the temperature in Kelvin.

For monatomic gases, $C_v = 3R/2$, and for diatomic gases, $C_v = 5R/2$.

Option A: 2 moles of He at 300 K

- Helium is a monatomic gas, so $C_v = 3R/2$.

$$U_A = 2 \times (3R/2) \times 300 = 900R$$

Option B: 56 kg of nitrogen at 300 K

- Nitrogen is a diatomic gas, so $C_v = 5R/2$.
- Molar mass of N_2 is 28 g/mol, so number of moles (n) is:
 $n_B = 56000 \text{ g} / 28 \text{ g/mol} = 2000 \text{ moles}$
- Internal energy: $U_B = 2000 \times (5R/2) \times 300 = 1,500,000R$

Option C: 8 g of oxygen at 300 K

- Oxygen is a diatomic gas, so $C_v = 5R/2$.
- Molar mass of O_2 is 32 g/mol, so number of moles (n) is:
 $n_C = 8 \text{ g} / 32 \text{ g/mol} = 0.25 \text{ moles}$
- Internal energy: $U_C = 0.25 \times (5R/2) \times 300 = 93.75R$

Option D: 1000 moles of argon at 900 K

- Argon is a monatomic gas, so $C_v = 3R/2$.
- Internal energy: $U_D = 1000 \times (3R/2) \times 900 = 1,350,000R$

Therefore, the correct option is B.

NEST Based Problem

1.

In thermodynamics, a reversible process is characterized by the surroundings being always in equilibrium with the system. This implies that the process can be reversed with infinitesimally small changes to the system and surroundings, and there is no net change in the system or surroundings over the entire process.

Therefore, the correct option is C.

2.

$$\Delta U = 120 \text{ J}, Q = 120 \text{ J}$$

$$Q = C_v \Delta T$$

$$\Delta T = Q / C_v$$

$$\Delta T = \frac{120 \text{ J}}{20 \text{ J/K}} = 6 \text{ K}$$

Therefore, the correct option is B.

3.

$$u_{rms} = \sqrt{\frac{3kT}{m}}$$

$$\frac{u_{rms_A}}{u_{rms_B}} = \frac{\sqrt{\frac{3kT}{2m}}}{\sqrt{\frac{3kT}{m}}} = \sqrt{\frac{2}{1}} = \sqrt{2}$$

Therefore, the correct option is B.

4.

$$\Delta S = C_v \ln\left(\frac{T_2}{T_1}\right) + R \ln\left(\frac{V_2}{V_1}\right)$$

Temperature = constant

$$\Delta S = R \ln\left(\frac{V_2}{V_1}\right)$$

$$\Delta S = R \ln\left(\frac{P_1}{P_2}\right)$$

$$\Delta S = R \ln\left(\frac{300R}{1L \times 1atm}\right)$$

$$\Delta S = R \ln(24.6)$$

Therefore, the correct option is B.

5.

$$A(P_1, V_1, T_1) \rightarrow B(P_2, V_1, T_2)$$

$$\Delta H_{overall} = \Delta H_1 + \Delta H_2$$

$$\Delta H_1 = \Delta U_1 + (P_2 V_1 - P_1 V_1)$$

$$\Delta H_1 = C_v(T_2 - T_1) + (P_2 V_1 - P_1 V_1)$$

$$\Delta H_2 = \Delta U_2 + (P_3 V_2 - P_2 V_1)$$

$$\Delta H_2 = -P_3(V_2 - V_1) + (P_3 V_2 - P_2 V_1)$$

$$\Delta H_2 = (P_3 V_2 - P_2 V_1)$$

$$\Delta H_{overall} = C_v(T_2 - T_1) + (P_2 V_1 - P_1 V_1) + P_3 V_2 - P_2 V_1$$

$$\Delta H_{overall} = C_v(T_2 - T_1) + P_3 V_1 - P_1 V_1$$

Therefore, the correct option is C.

6.

$$\Delta H = Q_p$$

Since H is a state function, ΔH will remain the same from both path isobaric and non-isobaric. But $q = q_p$ only when the path was isobaric.

$\Delta H = q_p$ only for the isobaric path.

Therefore, the correct option is A.

7.

$$Q_{rev} = 1000 \text{ J}, Q_{irrev} = 1200 \text{ J}$$

For reversible process:

$$\Delta S_{reversible} = \frac{1000 \text{ J}}{T}$$

For irreversible process:

$$\Delta S_{irreversible} = \frac{1200 \text{ J}}{T}$$

Since T is the same for both processes, we can compare the ratios:

$$\frac{\Delta S_{reversible}}{\Delta S_{irreversible}} = \frac{\frac{1000}{T}}{\frac{1200}{T}} = \frac{1000}{1200} = \frac{5}{6}$$

Therefore, the correct option is C.

8.

Heat lost copper = heat gain by gold

$$30 \times 0.385(318 - T) = 15 \times 0.129(T - 298)$$

Final temperature, $T = 315.1 \text{ K}$

$$T = 42.1^\circ\text{C}$$

Therefore, the correct option is A.

9.



$$\text{Moles of O}_2 \text{ used per day} = \frac{640}{32} = 20$$

$$\therefore \text{Moles of sucrose used per day or for 20 mole O}_2 = \frac{20}{12} \text{ mol} = \frac{20}{12} \times 342 \text{ g} = 570 \text{ g}$$

$$\text{Also, } \Delta H = \frac{5645 \times 20}{12} = 9408.34 \text{ kJ}$$

Therefore, the correct option is D.

10.

$$\Delta_r H = 57 \times \Delta H_{\text{CO}_2(g)} + 52 \times \Delta H_{\text{H}_2\text{O}(l)} - \Delta H_{\text{C}_{57}\text{H}_{107}\text{O}_6}$$

$$\Delta_r H = -14861.6 - 22429.5 - 70870 = 33578.9 \text{ kJ/mol}$$

Molar mass of $\text{C}_{57}\text{H}_{107}\text{O}_6 = 888.402 \text{ g/mol}$

When 888.402 g of $\text{C}_{57}\text{H}_{107}\text{O}_6$ is metabolized, $\Delta_r H = 33578.9 \text{ kJ}$

Therefore, when 1g of $\text{C}_{57}\text{H}_{107}\text{O}_6$ is metabolized

$$\Delta_r H = \frac{33578.9}{888.402} = 37.98 \text{ kJ}$$

Therefore, the correct option is C.

11.

$$\Delta_r H = \left[4 \times \Delta H_{\text{NO}} + 6 \times \Delta H_{\text{H}_2\text{O}} \right] - \left[4 \times \Delta H_{\text{NH}_3} + 5 \times \Delta H_{\text{H}_2\text{O}} \right]$$

$$= [4 \times 90 + 6 \times [-241.8]] - [4 \times [-46.1] + 5 \times 0]$$

$$= 180 - 8 + 1440.8 + [264.4]$$

$$= -905.6 \text{ kJ}$$

4 mole of $\text{NH}_3 \rightarrow 4 \times 14 \rightarrow 68 \text{ g of NH}_3 \rightarrow -905.6 \text{ kJ}$

$$\Rightarrow \frac{-905.6 \times 3}{68} = -39.9 \text{ kJ}$$

Therefore, the correct option is C.

12.

Applying Hess's law,

$$\Delta_r H = -3 \times \Delta H_1 + \Delta H_2 + 2 \times \Delta H_3 + 3 \times \Delta H_4$$

$$\Delta_r H = [3(110.5) - 282.9 + 2(-285.8) + 3(-74.8)]$$

$$\Delta_r H = -747.5 \text{ kJ}$$

Therefore, the correct option is C.

13.

According to question, the Enthalpy of the reaction will be the addition of both ΔH_1 and ΔH_2

$$\Delta H_{rxn} = \Delta H_1 + \Delta H_2$$

According to the question,

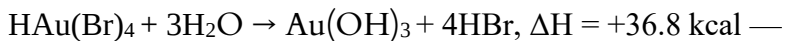
$$\Delta H_{rxn} = \frac{8 \times \Delta H_1 + 1 \times \Delta H_2}{9}$$

$$\Delta H_{rxn} = \frac{8 \times 45.5 + 1 \times 68.91}{9}$$

$$\Delta H_{rxn} = 48.137 \text{ kJ}$$

Therefore, the correct option is B.

14.



When 100% conversion occurs, heat absorbed = 8.8 kcal

$$\% \text{ conversion} = \frac{0.44}{8.8} \times 100 = 5$$

Therefore, the correct option is C.

15.

$$\begin{aligned}
 & \text{C}_3\text{H}_6 + \frac{9}{2} \text{O}_2 \rightarrow 3\text{CO}_2 + 3\text{H}_2\text{O} \\
 & 3\text{C} + 3\text{H}_2 \rightarrow \text{C}_3\text{H}_6 \quad \Delta H = 20.6 \text{ kJ mol}^{-1} \\
 & \text{C} + \text{O}_2 \rightarrow \text{CO}_2 \quad \Delta H = -394 \text{ kJ mol}^{-1} \\
 & \text{H}_2 + \frac{1}{2} \text{O}_2 \rightarrow \text{H}_2\text{O} \quad \Delta H = -394 \text{ kJ mol}^{-1} \\
 & (\Delta H_c)_{\text{C}_3\text{H}_6} = \left[3\Delta H_{\text{CO}_2} + 3\Delta H_{f(\text{H}_2\text{O})} - \Delta H_{f(\text{C}_3\text{H}_6)} \right] \\
 & \quad = [3 \times (-394) - 3(285.8) - 20.6] \\
 & (\Delta H_c)_{\text{C}_3\text{H}_6} = -2060 \text{ kJ mol}^{-1}
 \end{aligned}$$

Therefore, the correct option is C.

16.

The relation between internal energy and temperature for an ideal gas is given by the equation,

$$U = nC_vT.$$

If U decreases, and the number of moles and C_v remains constant, then T must also decrease.

For an adiabatic process, $\Delta S = 0$ (no heat exchange). Entropy remains constant during an adiabatic process.

Therefore, the correct option is A.

17.

Bond dissociation and bond energy will be constant for the Hydrogen molecule as it has only one type of bond.

Therefore, the correct option is A.

18.

$$\begin{aligned}
 & \text{C}_2\text{H}_5\text{--S--C}_2\text{H}_5 + \text{S} \rightarrow \text{C}_2\text{H}_5\text{--S--S--C}_2\text{H}_5 \\
 & \Delta H_f^0(\text{C}_2\text{H}_5\text{--S--C}_2\text{H}_5) = \Delta H_f^0(\text{C}_2\text{H}_5\text{--S--S--C}_2\text{H}_5) - \Delta H_{\text{S--S}} - \Delta H_{\text{sub}}\text{S} \\
 & \Delta H_{\text{S--S}} = 277.5 \text{ kJ/mol}
 \end{aligned}$$

Therefore, the correct option is B.

19.

This experiment is described by the Joule-Thomson effect, which is a phenomenon

related to the expansion of gases. According to the Joule-Thomson effect, when a gas expands into a vacuum or another region of lower pressure without doing any external work, its temperature remains constant. "The result is applicable to both ideal and real gases." This statement is correct. The Joule-Thomson effect is a general phenomenon applicable to both ideal and real gases under certain conditions. It doesn't depend on the specific nature of the gas and is a result of intermolecular interactions.

Therefore, the correct option is D.

20.

Isothermal refers to a process occurring at constant temperature. Free expansion occurs when a gas expands into a vacuum or an area of extremely low pressure without doing any work on the surroundings or receiving any work from the surroundings. In this case, the container is perfectly insulated, indicating no heat exchange with the surroundings. The sudden expansion results in no change in temperature, and since it's a free expansion, there is no work done on or by the system.

Therefore, the correct option is D.

21.

$$W = -nRT \ln\left(\frac{V_f}{V_i}\right)$$

$$PV = nRT$$

$$n = \frac{PV}{RT}$$

$$W = -\left(\frac{PV}{RT}\right)RT \ln\left(\frac{V_f}{V_i}\right)$$

$$W = -PV \ln\left(\frac{V_f}{V_i}\right)$$

$$W = -150 \text{ J}, P = 3 \text{ atm}, V_i = 2.50 \text{ L}$$

$$150 = (3 \times 2.50) \ln\left(\frac{V_f}{2.50}\right)$$

Solving for $\frac{V_f}{2.50}$

$$\ln\left(\frac{V_f}{2.50}\right) = \frac{150}{7.50}$$

Take exponential on both sides:

$$\frac{V_f}{2.50} = e^{\frac{150}{7.50}}$$

$$V_f = 2.50 \times e^{\frac{150}{7.50}}$$

$$V_f \approx 3.33 \text{ L}$$

Therefore, the correct option is A.

22.

$$\Delta S = nR \log \frac{P_1}{P_2}$$

for isothermal process

$$= 3 \times 8.314 \log \frac{2}{1}$$

$$= 7.508 \text{ J/K}$$

Therefore, the correct option is A.

23.

To find the work done during the expansion from V_1 to V_2 , we need to integrate the expression for pressure with respect to volume. The given equation is:

$$P = a/V + b$$

The work done, W , during an expansion is given by:

$$W = \int P \, dV$$

Substituting the expression for P :

$$W = \int (a/V + b) \, dV$$

Breaking this integral into two parts:

$$W = \int (a/V) \, dV + \int b \, dV$$

Calculating the first integral:

$$\int (a/V) \, dV = a [\ln(V)] \text{ from } V_1 \text{ to } V_2 = a \ln(V_2/V_1)$$

Calculating the second integral:

$$\int b \, dV = b [V] \text{ from } V_1 \text{ to } V_2 = b (V_2 - V_1)$$

Adding both parts, the total work done is:

$$W = a \ln(V_2/V_1) + b (V_2 - V_1)$$

Therefore, the correct option is D.

24.

To find the temperature rise experienced by the body, we use the formula:

$$Q = mc\Delta T$$

Where:

Q is the amount of heat produced,

m is the mass of the system,

c is the specific heat capacity,

ΔT is the temperature rise.

Given:

$$Q = 10 \text{ MJ} = 10 \times 10^6 \text{ J},$$

$$m = 8 \text{ kg},$$

$$c = 4.2 \text{ J/kg}\cdot\text{K}$$

Plugging in the values:

$$\Delta T = Q / (mc)$$

$$\Delta T = (10 \times 10^6 \text{ J}) / (8 \text{ kg} \times 4.2 \text{ J/kg}\cdot\text{K})$$

$$\Delta T = 10^7 / 33.6$$

$$\Delta T \approx 297.6 \text{ K}$$

Therefore, the correct option is B.

25.

Calculate the heat lost by the liquid water:

The water cools from 20°C to 0°C. The heat lost by the water is given by:

$$Q_{\text{lost}} = mc\Delta T$$

Where:

$$m = 500 \text{ g} = 0.5 \text{ kg},$$

$$c = 4.2 \text{ J/g}\cdot\text{K} \text{ (converted from } 75.6 \text{ J/Kmol)},$$

$$\Delta T = 20^\circ\text{C}.$$

Calculating Q_{lost} :

$$Q_{\text{lost}} = 500 \text{ g} \times 4.2 \text{ J/g}\cdot\text{K} \times 20 \text{ K} = 42,000 \text{ J}$$

Calculate the heat gained by the ice cubes:

Each ice cube contains 9.0 g of water, which must first melt at 0°C. The heat required to melt one ice cube is given by:

$$Q_{\text{gain (fusion)}} = n \Delta H_{\text{fusion}}$$

Where:

n is the number of moles of ice,

$$\Delta H_{\text{fusion}} = 6 \text{ kJ/mol.}$$

For 9 g of ice:

$$\text{Molar mass of water} = 18 \text{ g/mol, so } n = 9 \text{ g} / 18 \text{ g/mol} = 0.5 \text{ mol}$$

$$Q_{\text{gain (fusion)}} = 0.5 \text{ mol} \times 6 \text{ kJ/mol} = 3,000 \text{ J}$$

Determine the number of ice cubes needed:

To cool 500 g of water from 20°C to 0°C, the total heat absorbed by the ice cubes should equal the heat lost by the water:

$$\text{Number of ice cubes} = Q_{\text{lost}} / Q_{\text{gain (fusion)}}$$

$$\text{Number of ice cubes} = 42,000 \text{ J} / 3,000 \text{ J} = 14$$

Therefore, the correct option is C.

26.

Calculate the initial internal energy of the gases.:

The initial internal energy of each chamber is given by:

$$U_1 = (3/2) P_1 V_1$$

$$U_2 = (3/2) P_2 V_2$$

where P_1 , V_1 , and T_1 represent the pressure, volume, and temperature of the first chamber, and P_2 , V_2 , and T_2 represent the same for the second chamber.

Calculate the final internal energy after mixing:

After removing the partition, the gases mix and reach a final equilibrium temperature T_f . The total internal energy after mixing is given by:

$$U_{\text{final}} = (3/2) (P_1 V_1 + P_2 V_2) T_f$$

Apply the principle of energy conservation.

The total internal energy before and after removing the partition must be equal, so:

$$U_1 + U_2 = U_{\text{final}}$$

Substituting the expressions for U_1 , U_2 , and U_{final} :

$$P_1 V_1 T_1 + P_2 V_2 T_2 = (P_1 V_1 + P_2 V_2) T_f$$

Solving for T_f gives:

$$T_f = [T_1 T_2 (P_1 V_1 + P_2 V_2)] / [P_1 V_2 T_2 + P_2 V_1 T_1]$$

The final equilibrium temperature is:

$$T_f = [T_1 T_2 (P_1 V_1 + P_2 V_2)] / [P_1 V_2 T_2 + P_2 V_1 T_1]$$

Therefore, the correct option is C.

27.

The process is an isothermal expansion of n moles of an ideal gas.

The gas expands from an initial pressure P atm to a final pressure of 1 atm.

The expansion occurs in a number of steps, where the external pressure decreases by 1 atm in each step.

Work Calculation:

The work done in each step of the expansion is given by:

$$w_i = -nRT \ln \left(\frac{V_{i+1}}{V_i} \right)$$

Using the pressure relationship, the volume ratio in each step becomes proportional to the inverse of pressure:

$$\frac{V_{i+1}}{V_i} = \frac{P_i}{P_{i+1}}$$

Here, the external pressure in each step is $(P + 1 - i)$ atm.

Total Work:

Summing over all steps from $i = 1$ to $P - 1$, we have:

$$w = -nRT \sum_{i=1}^{P-1} \left(\frac{1}{P+1-i} \right)$$

Therefore, the correct option is A.

28.

For an isothermal process, the temperature of the system remains constant. As a result, the change in the system's internal energy (ΔU) is zero because an ideal gas's internal energy depends only on temperature.

Explanation:

For a triatomic nonlinear gas heated isothermally:

The heat added to the system (Q) is entirely used to work (W) against the surroundings, with no change in internal energy.

Mathematically, $\Delta U = 0$ since $\Delta U = Q - W$, and in an isothermal process for an ideal gas, ΔU is zero.

Thus, 0% of the heat energy is used to increase the internal energy, making the correct answer:

Therefore, the correct option is A.

29.

The relationship between pressure P and volume V in a polytropic process is:

$$PV^n = \text{constant}$$

Initial and final conditions are given as:

$$V_{\text{final}} = 4 \times V_{\text{initial}}$$

$$P_{\text{final}} = \frac{P_{\text{initial}}}{8}$$

Using the polytropic relation:

$$P_{\text{initial}} \cdot V_{\text{initial}}^n = P_{\text{final}} \cdot V_{\text{final}}^n$$

Substituting the values:

$$P_{\text{initial}} \cdot V_{\text{initial}}^n = \left(\frac{P_{\text{initial}}}{8} \right) \cdot (4 \cdot V_{\text{initial}})^n$$

Simplifying further:

$$1 = \frac{1}{8} \cdot 4^n$$

$$8 = 4^n$$

$$4^{1.5} = 4^n$$

n=1.5.

The formula for molar heat capacity C in a polytropic process is:

$$C = C_v + \frac{R}{1 - n}$$

Substituting the Values

$$C = \frac{3R}{2} + \frac{R}{1 - 1.5}$$

$$C = \frac{3R}{2} - 2R$$

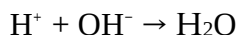
$$C = -4.2 \text{ J/K-mol}$$

Therefore, the correct option is A.

Exercise: 2

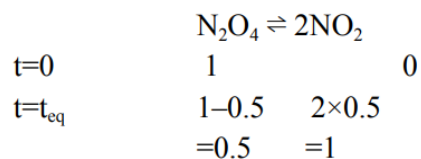
IAT PYQs

1. The heat of neutralization is primarily due to the formation of water (H₂O) from the reaction between hydrogen ions (H⁺) from the acid and hydroxide ions (OH⁻) from the base.



Therefore, the correct option is A.

2.



$\therefore$ Total number of moles = $0.5+1=1.5$
mol

$$P_{\text{N}_2\text{O}_4} = \frac{0.5}{1.5} \times 1 \text{ atm} = \frac{1}{3} \text{ atm}$$

$$P_{\text{NO}_2} = \frac{1}{1.5} \times 1 \text{ atm} = \frac{2}{3} \text{ atm}$$

$$K_P = \frac{P_{\text{NO}_2}^2}{P_{\text{N}_2\text{O}_4}} = \frac{(2/3)^2}{(1/3)} = 1.33 \text{ atm}$$

$$\Delta G^\circ = -2.303 \times 8.314 \times 333 \times \log(1.33) = -790 \text{ J/mol}$$

Therefore, the correct option is B.

3.

For an ideal gas, the relationship between the heat capacity at constant pressure C_p and the heat capacity at constant volume C_v is given by the well-known Mayer's relation:

$$C_p - C_v = R.$$

This relation is derived from the fact that at constant pressure, additional work is done by the gas to expand against the external pressure, whereas, at constant volume, no expansion work is done.

Therefore, the correct option is D.

4.

The given reaction represents the complete dissociation of methane into carbon and hydrogen atoms in the gaseous state. The enthalpy change (ΔH°) corresponds to the total energy required to break all four C-H bonds in methane.

Option A: ΔH° is the mean bond enthalpy of a C-H bond. This statement is FALSE because ΔH° here represents the total energy required to break all four C-H bonds in methane, not the mean bond enthalpy. The mean bond enthalpy is calculated by dividing the total enthalpy change by the number of bonds, i.e., mean bond enthalpy = $1665/4 \text{ kJ mol}^{-1}$.

Option B: All four C-H bonds in CH_4 are identical in bond length and energy. This statement is TRUE, as methane is a symmetrical molecule with identical C-H bonds.

Option C: The energy required to break individual C-H bonds in successive steps is different. This statement is TRUE because the bond dissociation energy increases with each successive bond-breaking due to changes in the electronic environment around the central carbon atom.

Option D: Mean C-H bond enthalpies differ slightly from compound to compound. This statement is TRUE because the mean bond enthalpy depends on the molecular environment, which varies between different compounds.

Therefore, the correct option is A.

5.

This is an exothermic reaction, which means that the enthalpy change (ΔH) is negative ($\Delta H < 0$). Additionally, the reaction produces gases (C(g) and D(g)), indicating an increase in entropy ($\Delta S > 0$).

The Gibbs free energy change (ΔG) for a reaction is given by the equation:

$$\Delta G = \Delta H - T\Delta S$$

For a reaction to be spontaneous, ΔG must be negative. In this case:

- ΔH is negative (exothermic reaction).
- ΔS is positive (due to the production of gases).

Since both terms contribute to making ΔG negative, the reaction will be spontaneous at all temperatures.

Therefore, the correct option is A.

NEST PYQ's

1.

Reversible Isothermal Expansion (X to Y): In an isothermal process, the temperature of the gas remains constant. A horizontal line represents the TS (Temperature vs. Entropy) graph as temperature remains constant and entropy increases. In the UT (Internal Energy vs. Temperature) graph, internal energy does not change, as internal energy remains constant during isothermal processes for an ideal gas.

Reversible Adiabatic Expansion (X to Z): In an adiabatic process, there is no heat exchange, so the entropy of the gas remains constant. The TS graph is represented by a vertical line, as there is no change in entropy during the adiabatic process. In the UT graph, internal energy

decreases as temperature decreases, which aligns with the characteristics of an adiabatic expansion.

The correct graphs that represent the processes are:

1. Graph B: The TS graph correctly shows a horizontal line for isothermal expansion and a vertical line for adiabatic expansion.
2. Graph D: The UT graph correctly shows no change in internal energy during isothermal expansion and a decrease in internal energy during adiabatic expansion.

Therefore, the correct answer is B and D.

2.

Option A: If ΔH is negative, but ΔS is positive, the reaction will never take place. This is incorrect. A reaction with negative enthalpy change ($\Delta H < 0$) and positive entropy change ($\Delta S > 0$) is typically spontaneous at all temperatures because it results in a negative Gibbs free energy (ΔG). The Gibbs free energy is given by $\Delta G = \Delta H - T\Delta S$, which becomes more negative as both ΔH is negative and ΔS is positive.

Option B: If both ΔH and ΔS are positive, the reaction will always be spontaneous. This is incorrect. A reaction with both positive enthalpy ($\Delta H > 0$) and positive entropy ($\Delta S > 0$) will only be spontaneous at high temperatures, where the entropy term $T\Delta S$ dominates the enthalpy term in the Gibbs free energy equation ($\Delta G = \Delta H - T\Delta S$).

Option C: If ΔG° for a reaction is positive, the reaction can still take place under suitable conditions. This is correct. Even if the standard Gibbs free energy (ΔG°) is positive, the reaction can still occur under non-standard conditions (e.g., changes in pressure, temperature, or concentration), which can alter the Gibbs free energy to a negative value, making the reaction feasible.

Option D: If ΔG° for the forward reaction is negative, only the forward reaction will take place, and such reactions can never be reversible. This is incorrect. Even if the Gibbs free energy (ΔG°) for the forward reaction is negative, the reverse reaction can still occur, though it will be less likely. Additionally, many reactions with negative ΔG° are reversible, as they can reach an equilibrium state.

Therefore, the correct answer is C.

3.

Heating a solid: Initially, as the solid is heated, the temperature increases linearly with time.

Melting point: When the solid reaches its melting point, the temperature remains constant during the phase change (melting), as all the heat energy is used to break the intermolecular bonds rather than raising the temperature. This results in a horizontal line in the T vs. t graph.

Heating the liquid: After the solid has completely melted, the temperature of the liquid starts to increase linearly again with further heating.

The complete T vs. t plot should have a linear increase, followed by a plateau (constant temperature during melting), and then another linear increase after melting. This corresponds to Option B.

Therefore, the correct answer is C.

4.

Isothermal Process: In an isothermal expansion, the temperature remains constant. For an ideal gas, the internal energy change (ΔE) depends only on temperature. Since the temperature remains constant, $\Delta E = 0$. The entropy change (ΔS) is positive because the gas expands, increasing the disorder in the system.

$$\Delta E = 0, \Delta S > 0.$$

Therefore, the correct answer is B.

5.

At the boiling point (100°C for water at 1 atm), the system is at equilibrium during vaporization. The Gibbs free energy change (ΔG) is given by $\Delta G = \Delta H - T\Delta S$. At equilibrium, $\Delta G = 0$ since the liquid and vapour phases coexist at the boiling point.

Therefore, the correct answer is B.

6.

Curie's Law states that the molar magnetic susceptibility (χ) of a paramagnetic material is given by $\chi = C/T$, where C is the Curie constant and T is the absolute temperature. Taking the reciprocal of χ , we get $\chi^{-1} = T/C$. This equation indicates that χ^{-1} is directly proportional to T, resulting in a linear plot of χ^{-1} vs T with a positive slope.

Therefore, the correct answer is B.

7.

The relation between the standard Gibbs free energy change (ΔG°) and the equilibrium constant (K_p) is given by $\Delta G^\circ = -RT \ln K_p$.

Rearranging for K_p , we get $K_p = \exp(-\Delta G^\circ/RT)$.

Given, $\Delta G^\circ = 31.85 \text{ kJ/mol} = 31,850 \text{ J/mol}$, and $T = 298 \text{ K}$.

Substituting these values into the equation for K_p gives

$$K_p = \exp(-31,850 / (8.314 \times 298)).$$

The vapour pressure of mercury is equal to K_p in terms of partial pressure

Therefore, the correct answer is B.

8.

Entropy Change of Surroundings (ΔS_{surr}): Since the surroundings do not perform any work, the entropy change of the surroundings is given by:

$$\Delta S_{\text{surr}} = -q_{\text{irr}} / 300.$$

This matches Option A.

Entropy Change of the System (ΔS_{sys}): For an ideal gas undergoing expansion, the entropy change of the system is:

$$\Delta S_{\text{sys}} = nR \ln(V_2/V_1).$$

Since pressure changes from 2 atm to 1 atm at a constant temperature, we get:

$$\Delta S_{\text{sys}} = R \ln(2).$$

This matches Option B.

Relation between ΔS_{sys} and q_{irr} : In this problem, the relationship $\Delta S_{\text{sys}} = q_{\text{irr}} / 300$ is not necessarily true, as q_{irr} involves irreversible heat transfer, which does not directly equate to the entropy change of the system.

Heat Transfer (q_{irr}): We know that $q_{\text{irr}} < 300 R \ln 2$, which matches Option D.

The correct answers are Options A, B, and D.

9.

For the process $P \rightarrow Q$, which is adiabatic and irreversible, the entropy change $\Delta S_{PQ} > 0$.
For the process $P \rightarrow R$, which is adiabatic and reversible, the entropy change $\Delta S_{PR} = 0$.
For the process $Q \rightarrow R$, to ensure the total entropy change aligns with $\Delta S_{PR} = 0$, the entropy change ΔS_{QR} must be negative.

The correct answer is Option C.

10.

Option A: All processes are adiabatic, meaning no heat exchange occurs during the cycle. This makes it challenging to form a complete cycle, as a combination of only adiabatic processes is theoretically invalid for a cyclic process. Therefore, this combination cannot lead to a cyclic process, making it a correct answer.

Options B, C, and D involve combinations of adiabatic and isothermal processes, making them feasible for forming a cyclic process.

The correct answer is Option A.

11.

Statement A: Work done is zero for process 3.

If process 3 involves constant volume, the work done is indeed zero. Therefore, this statement is true.

Statement B: Heat is completely converted into work for process 1.

If process 1 is isothermal, the heat absorbed is fully converted into work, making this statement true.

Statement C: Entropy remains constant in process 2.

We cannot confirm this statement without explicit information that process 2 is adiabatic, making it false.

Statement D: The average speed of molecules remains constant for process 3.

This cannot be validated without knowing if the temperature is constant, making this statement false.

The correct answers are Options A and B.

12.

Initial Conditions:

Compartment 1: $T_1 = 400\text{ K}$, $V_1 = 2\text{ L}$, containing 1 mole of an ideal gas.

Compartment 2: $T_2 = 200\text{ K}$, $V_2 = 4\text{ L}$, containing 1 mole of an ideal gas.

The compartments are separated by a diathermal piston, which allows heat exchange between them.

When the piston is released:

- The gases in the two compartments interact thermally through the diathermal piston, leading to a redistribution of energy and volume.

- The system evolves towards thermodynamic equilibrium, where the temperature and pressure become uniform in both compartments.

Graph Analysis:

- Graph C:

- This graph shows the pressure-equalizing between the two compartments over time, which is expected in a spontaneous process.

- Initially, there is a difference in pressure between the compartments, but as the system reaches equilibrium, the pressures stabilize at the same value.

- Graph D:

- This graph depicts the change in volume in each compartment. The oscillatory behaviour in volume indicates adjustments before stabilizing at equilibrium.

- This behaviour is possible due to the movement of the piston as the system seeks equilibrium.

The correct answers are Options C and D

13.

Understanding the Path Descriptions:

- Path 1 (Reversible Process): In a reversible process, the system changes state infinitesimally, maintaining equilibrium throughout.

- Path 2 (Irreversible Process): In an irreversible process, finite deviations from equilibrium lead to higher work done due to internal resistance.

Work Done Comparison:

- The relation given in the problem, $dw_{\text{rev}} < dw_{\text{irrev}}$, indicates that the work done in the reversible process is less than that in the irreversible process.

- This is consistent with the fact that irreversible processes generally require more work to overcome dissipative forces.

Heat Transfer and Entropy Change:

- In a reversible process, the entropy change (dS_{rev}) is minimal, while in an irreversible process, the entropy change (dS_{irrev}) is greater due to the generation of additional entropy.

- However, heat transfer is generally greater in the reversible process than in the irreversible one, i.e., $dq_{\text{rev}} > dq_{\text{irrev}}$.

Energy Conservation Relation:

- The total energy change must be the same for both paths, as they connect the same initial and final states.
- Therefore, the relation $dw_{\text{rev}} + dq_{\text{rev}} = dw_{\text{irrev}} + dq_{\text{irrev}}$ correctly represents the total energy balance for the two processes.

The correct answer is Option C.

14.

Given:

- P→Q: Isothermal and irreversible.
- Q→R: Adiabatic and reversible.
- R→P: Unknown, but part of the cyclic process.

Statement A:

- The relation $q_1/T_1 + q_2/T_2 = 0$ comes from the Clausius inequality, which applies to reversible processes.
- Since P→Q is irreversible, this relation does not hold for the given process.
- Therefore, Statement A is correct.

Statement B:

- In a cyclic process, the total energy change is zero, but the magnitudes of q_1 and q_2 do not need to be the same.
- Different amounts of heat can be transferred during the different legs of the cycle.
- Therefore, Statement B is incorrect.

Statement C:

- The entropy change for the reversible adiabatic process R→P is given by:
 $\Delta S = nR \ln(V_1/V_2)$, where V_1 and V_2 are the respective volumes.
- This is the correct expression for entropy change in an adiabatic reversible process.
- Therefore, Statement C is correct.

Statement D:

- In the cyclic process P→Q→R→P, the total entropy change must be zero.
- However, it is possible for the segment R→P to have a negative entropy change ($\Delta S < 0$), as long as the overall cycle maintains zero total entropy change.
- Therefore, the cyclic process is possible, making Statement D correct.

The correct answers are Options A, C, and D.

15.

1. Given Conditions:

- The reaction occurs in an insulated, sealed container.
- Temperature increases from 200 K to 400 K, indicating an exothermic reaction.
- The pressure remains constant, so the total number of moles should ideally decrease to maintain constant pressure.

2. Checking Option C:

- Reaction: $X(g) + Y_2(g) \rightarrow XY_2(g)$
- Number of Moles:
 - Reactants: 2 moles (1 mole of X and 1 mole of Y_2).
 - Products: 1 mole (1 mole of XY_2).
- The total number of moles decreases from 2 to 1, which aligns with maintaining constant pressure when the temperature increases.

3. Conclusion:

- The reaction in Option C matches the given conditions of constant pressure and an increase in temperature, making it the correct answer.

The correct answer is Option C

16.

Calculating Enthalpy Change (ΔH):

- For a monatomic ideal gas, $\Delta H = n C_p \Delta T$.
- $C_p = (5/2) R$, $\Delta T = T_2 - T_1 = 420 \text{ K} - 300 \text{ K} = 120 \text{ K}$.
- Substituting the values:
$$\Delta H = 1 \times (5/2) \times 8.314 \times 120 \approx 2494 \text{ J}.$$
- Converting to L atm: $1 \text{ L atm} \approx 101.3 \text{ J}$, so $\Delta H \approx 1 \text{ L atm}$.

Calculating Entropy Change (ΔS):

- The total entropy change is given by:
$$\Delta S = n C_v \ln(T_2/T_1) + nR \ln(V_2/V_1), \text{ where } C_v = (3/2) R.$$
- Substituting the values:
$$\Delta S = 8.314 [(3/2) \ln(1.4) + \ln(10)] \text{ J/K}.$$
- Converting to L atm K^{-1} :

$$\Delta S = (3 \ln(1.4) + 2 \ln(10))/600 \text{ L atm K}^{-1}.$$

Conclusion:

- The enthalpy change ΔH is 1 L atm, and the entropy change ΔS is $(3 \ln(1.4) + 2 \ln(10))/600 \text{ L atm K}^{-1}$.

Therefore, the correct answer is Option D.

17.

Analyzing Statement D:

- For an isothermal expansion of an ideal gas, the entropy change (ΔS) is given by:

$$\Delta S = nR \ln(V_2/V_1).$$
- If the final volume (V_2) is double the initial volume (V_1), then:

$$\Delta S = nR \ln(2).$$
- This entropy change is independent of temperature as it only depends on the ratio of volumes and the number of moles.
- Therefore, Statement D is correct.

Why the Other Statements Are Incorrect:

- Statement A: Entropy change for the system is the same whether the process is reversible or irreversible.
- Statement B: An irreversible adiabatic process can generate entropy, making it nonzero.
- Statement C: In an isothermal process, heat exchange does occur to maintain a constant temperature, contradicting the statement.

The correct answer is Option D.

18.

Understanding the Pressure-Density Plot:

- The given plot shows a cyclic path from X to Y to Z and back to X.
- For an ideal gas, the relation $P = \rho RT$ applies, indicating a direct relationship between pressure, density, and temperature.

Analyzing Option A (Temperature vs. Density - Curved Path):

- This plot represents a curved path where temperature decreases as density increases.
- It correctly reflects the behaviour of an ideal gas undergoing such a transformation, matching the cyclic process.
- Therefore, Option A is correct.

Analyzing Option B (Temperature vs. Density - Linear Path):

- This plot shows a linear relationship between temperature and density.
- However, the original pressure-density plot suggests a potentially non-linear transformation, making this option inconsistent.
- Therefore, Option B is incorrect.

Analyzing Option C (Volume vs. Temperature - Linear Path):

- This plot shows a linear increase in volume with temperature.
- This matches the behaviour of the given cyclic process, where volume and temperature can increase simultaneously.
- Therefore, Option C is correct.

Analyzing Option D (Temperature vs. Pressure - Linear Path):

- This plot shows a linear relationship between temperature and pressure, consistent with the ideal gas behaviour.
- It aligns with the given cyclic process, as pressure and temperature change linearly.
- Therefore, Option D is correct.

The correct answers are Options A, C, and D.

19.

Work Done by the Gas:

- The work done by the gas during expansion against a constant external pressure is given by:

$W = P_{\text{ext}} \times \Delta V$, where P_{ext} is the external pressure, and ΔV is the change in volume.

- Here, $P_{\text{ext}} = 1 \text{ atm}$ and $\Delta V = V_{\text{final}} - V_{\text{initial}} = 10 \text{ L} - 1 \text{ L} = 9 \text{ L}$.
- Therefore, $W = 1 \text{ atm} \times 9 \text{ L} = 9 \text{ L atm}$.
- This value does not match Option A (10 L atm), and the formula in Option B is not applicable for work calculation.

Entropy Change (ΔS):

- For an isothermal expansion, the change in entropy is calculated as:

$\Delta S = nR \ln(V_{\text{final}}/V_{\text{initial}})$, where R is the gas constant.

- Here, $R = 0.082 \text{ L atm K}^{-1}$, $V_{\text{initial}} = 1 \text{ L}$, and $V_{\text{final}} = 10 \text{ L}$.

- Substituting the values:

$\Delta S = n \times 0.082 \times \ln(10)$.

- Simplifying further:

$\Delta S \approx n \times 0.082 \times 2.303 \approx n \times (2.303/30) \text{ L atm K}^{-1}$.

The correct answer is Option D.

20.

Analyzing Statement A:

- Activation energy (E_a) represents the minimum energy required for reactants to transition to products.
- While E_a can be zero for some elementary reactions, it is not always the case for all reactions.
- Therefore, Statement A is incorrect.

Analyzing Statement B:

- For endothermic reactions, the activation energy (E_a) must be greater than or equal to the heat of the reaction (ΔH).
- The minimum possible value of E_a equals ΔH when the transition state energy matches the energy of the products.
- Therefore, Statement B is correct.

Analyzing Statement C:

- Although E_a is a kinetic parameter and ΔH is a thermodynamic parameter, they can still be related.
- In an endothermic reaction, E_a must account for both the energy barrier and the overall energy change (ΔH).
- Therefore, Statement C is incorrect.

Analyzing Statement D:

- For atom-atom reactions (elementary reactions), the activation energy (E_a) can be zero.
- This is because such reactions may proceed directly without an intermediate energy barrier.
- Therefore, Statement D is correct.

The correct answers are Options B and D.

21.

Change in Entropy (ΔS)

For an ideal gas, the entropy change can be described by the following general formula:

$$\Delta S = nC_v \ln \left(\frac{T_2}{T_1} \right) + nR \ln \left(\frac{V_2}{V_1} \right)$$

For a monatomic ideal gas.

$$C_v = \frac{3}{2}R$$

Deriving the Expression for ΔS .

1. **First Term:** Entropy change due to temperature change

$$\Delta S_T = nC_v \ln \left(\frac{T_2}{T_1} \right)$$

$$\Delta S_T = \frac{3}{2}nR \ln \left(\frac{T_2}{T_1} \right)$$

Second Term: Entropy change due to volume change

$$\Delta S_V = nR \ln \left(\frac{V_2}{V_1} \right)$$

Combining Both Terms

$$\Delta S = \frac{3}{2}nR \ln \left(\frac{T_2}{T_1} \right) + nR \ln \left(\frac{V_2}{V_1} \right)$$

If we factor the terms carefully, this expression aligns with the general form of ΔS , where the components represent the product of the initial and final states of pressure, volume, and temperature—incorporated in a modified format.

$$\Delta S = \frac{3}{2} \left(P_1 V_1 T_1 \ln \left(\frac{T_2}{T_1} \right) + P_2 V_2 T_2 \ln \left(\frac{V_2}{V_1} \right) \right)$$

The correct answer is D.

Adiabatic Expansion:

- In an adiabatic expansion, there is no heat exchange with the surroundings ($q = 0$), so all the work done by the gas is due to the change in internal energy.
- For a monatomic ideal gas, the internal energy change ΔU is given by:
$$\Delta U = \left(\frac{3}{2}\right) nR (T_2 - T_1),$$
 where T_1 and T_2 are initial and final temperatures.

Work Done in Adiabatic Expansion:

- The work done during adiabatic expansion is given by:
$$w = (C_v/R) V_1 (P_2 - P_1),$$
 where $C_v = 3R/2$ for a monatomic ideal gas.
- Substituting C_v into the equation:
$$w = (3R/2R) \times V_1 (P_2 - P_1) = (3/2) V_1 (P_2 - P_1).$$
- This matches the expression given in Option D, making it the correct answer.

Explanation of Other Options:

- Option A: Describes the entropy change in an isothermal expansion, which is irrelevant to adiabatic expansion.
- Option B: Correctly states that the entropy change is zero in adiabatic expansion, but the question specifically asks for the work done.
- Option C: Represents the work done in an isothermal process, which is not applicable here.

The correct answer is Option D

23.

Cyclic Process:

- In a cyclic process, the gas returns to its initial state, meaning the total change in internal energy (ΔU) over one complete cycle is zero.

Work Done in Process $Y \rightarrow Z$ (Option D):

- The process $Y \rightarrow Z$ is described as isochoric (constant volume).
- In an isochoric process, there is no change in volume ($\Delta V = 0$), so the work done is given by:
$$w = P \Delta V = 0.$$
- Therefore, the work done in process $Y \rightarrow Z$ is zero, making Option D correct.

Explanation of Other Options:

- Option A: In the process $X \rightarrow Y$, if it is isothermal, ΔU is zero, and ΔH is positive due to heat absorption.
- Option B: In the process $Z \rightarrow X$, if it is adiabatic, ΔH is zero, but ΔU is positive due to temperature increase.
- Option C: Correctly describes the work done in an isothermal compression from P_2 to P_3 at temperature T_2 .

The correct answer is Option D.

